

Surface Area and Scale Design Workshop

Answer Key

Grade 6–7 — Geometry

Quick Answers

1. 392	4. 52	7. 287.46	10. 10
2. 78	5. 84	8. 116	11. —
3. 207.35	6. 240	9. 3700	12. 113.10

Each solution names the formula, substitutes the givens, and keeps the unit through every line. Answers marked $\approx$ are rounded to two decimal places; accept equivalent exact forms such as 66π .

1. Closed carton, $\ell = 12$ cm, $w = 8$ cm, $h = 5$ cm. Total surface area?

Solution: A prism has three pairs of matching faces, so $SA = 2(\ell w + \ell h + wh)$.

$$\begin{aligned} SA &= 2(12 \cdot 8 + 12 \cdot 5 + 8 \cdot 5) \\ &= 2(96 + 60 + 40) = 2(196) \end{aligned}$$

392 cm² (Getting 196 means the pairs were never doubled.)

2. Scale 1 cm = 4 m; room drawn 6.5 cm by 3.25 cm. Actual perimeter?

Solution: Convert each drawing length first, because perimeter is a length. Actual sides: $6.5 \cdot 4 = 26$ m and $3.25 \cdot 4 = 13$ m. Then $P = 2(26 + 13) = 2(39)$. 78 m

3. Closed cylinder, $r = 3$ cm, $h = 8$ cm. Total surface area?

Solution: Curved side (a rolled-up rectangle) + two circles: $SA = 2\pi rh + 2\pi r^2 = 2\pi r(r + h)$.

$$SA = 2\pi(3)(3 + 8) = 66\pi \approx 207.3451 \dots$$

207.35 cm² (150.80 is the curved side alone — the ends were dropped.)

4. Workshop floor plan on a 1 m grid. Total floor area?

Solution: Cut the plan into two rectangles along the step. The lower rectangle is 8 by 5, giving 40; the upper left rectangle is 3 wide and 4 tall, giving 12. Total: $40 + 12 = 52$. (A shoelace count of the six corners gives the same value.) 52 m²

5. Shed 6 m by 4 m by 3 m; paint four walls and roof, not the floor.

Solution: Take the faces one kind at a time instead of using the closed-prism formula. Two long walls: $2(6 \cdot 3) = 36$. Two short walls: $2(4 \cdot 3) = 24$. Roof: $6 \cdot 4 = 24$. Floor: not painted. Total = $36 + 24 + 24$. 84 m² (108 is the full closed prism — the floor was painted by mistake.)

6. Scale 1 cm = 4 m; deck drawn 5 cm by 3 cm. Actual area?

Solution: Scale the two lengths, then multiply — never scale the area once. Actual sides: $5 \cdot 4 = 20$ m and $3 \cdot 4 = 12$ m, so $A = 20 \cdot 12$. 240 m^2
Equivalently: drawing area 15 cm^2 times $4^2 = 16$. The common error, $15 \cdot 4 = 60$, uses the length factor on an area.

7. Label wrapped on a can, $r = 4$ cm, $h = 11$ cm, with a 1 cm overlap.

Solution: Unrolled, the label is a rectangle. Its height is the can's height, 11 cm; its length is the circumference plus the overlap: $2\pi(4) + 1 = 25.1327 \dots + 1$.

$$A = (2\pi \cdot 4 + 1)(11) \approx 287.4601 \dots$$

287.46 cm^2 (276.46 leaves out the glued overlap.)

8. Two identical L-shaped panels: 10 m by 4 m with a 6 m by 3 m rectangle on top.

Solution: One panel splits into $10 \cdot 4 = 40$ and $6 \cdot 3 = 18$, so one panel is 58 m^2 . The design needs two of them: $2(40 + 18) = 2(58)$. 116 m^2 (58 answers for one panel only.)

9. Model at 1 : 5 has surface area 148 cm^2 . Real crate's surface area?

Solution: Every length is multiplied by 5, and an area is a length times a length, so area is multiplied by $5^2 = 25$: $148 \cdot 25$. 3700 cm^2
(740 uses the length factor 5 on an area. Volume would scale by 5^3 .)

10. Closed box, square base 8 cm by 8 cm, total surface area 448 cm^2 . Height?

Solution: Write the surface area in terms of h : two square ends of $8 \cdot 8$ and four rectangular sides of $8h$.

$$2(8 \cdot 8) + 4(8h) = 448$$

$$128 + 32h = 448$$

$$32h = 320$$

10 cm Check: $128 + 4(8)(10) = 128 + 320 = 448 \checkmark$

11. Design decision: cube 6 by 6 by 6 versus box 12 by 6 by 3, both of volume 216.

Model answer (open reasoning — review by hand): Box A (cube): $SA = 6(6 \cdot 6) = 216$ square centimetres. Box B: $SA = 2(12 \cdot 6 + 12 \cdot 3 + 6 \cdot 3) = 2(72 + 36 + 18) = 252$ square centimetres. Both hold 216 cubic centimetres, but B needs 36 square centimetres more cardboard, so Box A

uses less material. The general reason: for a fixed volume, the closer a box is to a cube, the smaller its surface area; a long flat box has more outside for the same inside. A designer might still choose B — flat boxes stack and ship better on a pallet, they fit a wider product, and they give a larger printable front face for the label. Award credit for any well-argued practical reason (shipping, shelf display, product shape, ease of packing) that is stated alongside the correct numerical comparison.

12. Tank drawing at 1 cm = 2.5 m: 1.6 cm across, 3.2 cm tall. Painted area (curved wall + top only)?

Solution: Scale first, then apply the surface-area formulas. Actual diameter = $1.6 \cdot 2.5 = 4$ m, so $r = 2$ m; actual height = $3.2 \cdot 2.5 = 8$ m.

$$\text{curved wall} = 2\pi rh = 2\pi(2)(8) = 32\pi$$

$$\text{top} = \pi r^2 = \pi(2)^2 = 4\pi$$

$$\text{painted} = 36\pi \approx 113.0973\dots$$

113.10 m^2 (125.66 paints both circles; the base is on the ground.)